E12 — Hessian Spectral Tracking## Experiment OverviewThis experiment tracks the **Hessian eigenvalue spectrum** during optimization to investigate whether Muon's spectral normalization leads to better-conditioned optimization landscapes compared to standard SGD.For the linear matrix sensing problem, the Hessian at any point X is:H = (2/m) * sum_i vec(A_i) @ vec(A_i)^Twhich is **independent of X** for the MSE loss. We track the condition number (ratio of max to min eigenvalue) at regular intervals to characterize the problem's intrinsic geometry.**Algorithms**: Muon-Exact, SGD**Configuration**: d=50, r=5, lr=0.01, track_every=50 steps, 5 random seeds

Scientific Question**Hypothesis**: Muon's spectral normalization may interact differently with the problem's Hessian structure, potentially leading to more favorable effective conditioning or faster progress in directions of high curvature.Although the exact Hessian is constant for linear matrix sensing, tracking its spectrum provides insight into:1. The intrinsic condition number of the problem2. Whether algorithm behavior correlates with Hessian properties3. How eigenvalue ratios evolve (or not) during optimization**Key Metrics**:- eigen_ratio_start: Condition number at optimization start- eigen_ratio_end: Condition number at

optimization end- eigen_ratio_min/max: Range of condition numbers observed- K_epsilon: Iterations to convergence

Experimental Design| Parameter | Value ||-----------|-------|| Problem | Matrix Sensing (MS) || Algorithms | Muon-Exact, SGD || Dimension d | 50 || Target rank r | 5 || Learning rate | 0.01 || Track interval | Every 50 iterations || Max iterations | 2000 || Random seeds | 5 (0-4) |The Hessian is explicitly constructed as H = (2/m) * sum_i vec(A_i) vec(A_i)^T in R^(d^2 x d^2) space, and its eigenvalues computed via eigvalsh().

Data Loading & Inspection

In [ ]:
```
import pandas as pdimport numpy as npimport matplotlib.pyplot as pltimport s
```

Exploratory Data Analysis### Hessian Condition Number Statistics

In [ ]:
```
hessian_cols = ['eigen_ratio_start', 'eigen_ratio_end', 'eigen_ratio_min', '
```

Convergence Comparison

In [ ]:
```
summary = df.groupby('algo').agg(    n=('seed', 'count'),    K_eps_mean=('K_
```

Visualizations

Plot 1: Hessian Condition Number Distribution

In [ ]:
```
fig, axes = plt.subplots(1, 2, figsize=(14, 6))algo_list = ['Muon-Exact', 'S
```

Plot 2: Condition Number vs Convergence SpeedWe examine whether the problem's condition number correlates with the number of iterations needed for convergence.

In [ ]:
```
fig, ax = plt.subplots(figsize=(10, 6))for algo, color, marker in [('Muon-Ex
```

Plot 3: Comprehensive Algorithm Comparison Dashboard

In [ ]:
```
fig, axes = plt.subplots(2, 2, figsize=(14, 12))algo_list = ['Muon-Exact', '
```

Statistical Tests### Paired t-test: Muon-Exact vs SGD

In [ ]:
```
from scipy.stats import ttest_rel, wilcoxonmuon_subset = df[df['algo'] == 'M
```

Conclusions & Interpretation### Key Findings1. **Hessian is Problem-Intrinsic**: For linear matrix sensing, the Hessian condition number is constant throughout optimization (start = end = min = max). The values are approximately ~2100, indicating a moderately ill-conditioned problem.2. **No Hessian Conditioning Advantage**: Since the Hessian is independent of the current iterate X for the linear MS problem, Muon and SGD operate on the *same* curvature landscape. Any performance differences must be attributed to the algorithms' update directions, not to differences in Hessian conditioning.3. **Muon Converges Faster**: Despite identical Hessian structure, Muon-Exact achieves lower K_epsilon (~41 iterations) compared to SGD (~47 iterations), representing a ~13% speedup in iteration count.4. **Solution Quality Trade-off**: As observed in E11, SGD achieves much better final loss values (~1e-32) compared to Muon (~5e-3), at the cost of slightly more iterations.### Implications- The performance difference between Muon and SGD on this problem is **not** due to differences in Hessian conditioning (since the Hessian is the same for both).- Muon's advantage comes from its **spectral normalization** of the update direction, which may provide more effective progress in certain spectral directions.- The constant Hessian condition number (~2100) is relatively large, suggesting the problem is moderately ill-conditioned, which may explain why spectral methods show benefits.- For problems where the Hessian *does* vary with X (non-convex settings like matrix factorization), Muon's effect on Hessian conditioning may be more pronounced and warrants further study.